

MARFA, TX, USA

WMO: 722640

Lat: 30.371N

Lon: 104.017W

Elev: 4849

StdP: 12.30

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93035

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.7	19.6	-1.9	5.9	32.0	0.9	6.9	35.5	32.6	54.8	28.6	52.9	5.1	350	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	30.8	94.5	60.9	91.4	61.1	90.2	61.1	67.5	80.5	66.6	79.8	65.8	79.3	10.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	107.2	69.9	63.3	104.4	69.6	62.7	102.1	69.6	35.0	80.7	34.2	80.3	33.4	78.6	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	23.8	20.3	8.9	98.6	4.0	2.3	6.0	100.3	3.6	101.6	1.3	102.9	-1.6	104.6
			7.5	69.3	3.9	1.4	4.7	70.3	2.5	71.1	0.3	71.9	-2.5	72.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	42.0	46.4	52.1	59.1	67.2	74.2	73.8	73.0	68.4	60.1	50.2	42.6
	DBStd	13.15	6.89	7.83	7.01	6.49	5.76	3.70	3.20	3.25	4.39	6.40	6.92	7.04
	HDD50	804	256	143	61	11	1	0	0	0	0	9	80	243
	HDD65	3202	713	520	399	193	43	1	1	1	19	172	445	695
	CDD50	4147	9	43	127	282	533	727	739	714	553	322	85	13
	CDD65	1071	0	0	1	15	110	278	275	250	122	20	0	0
	CDH74	14815	1	31	192	759	2321	3832	3003	2582	1457	585	52	0
	CDH80	6056	0	2	20	170	982	1970	1307	1037	454	112	2	0
(14) Wind	WSAvg	8.6	8.8	9.6	9.9	10.4	9.7	8.3	7.6	7.2	7.4	7.9	8.1	8.3
(15) Precipitation	PrecAvg	14.4	0.6	0.6	0.4	0.5	0.8	1.8	2.8	2.3	2.0	1.3	0.6	0.6
	PrecMax	28.1	1.8	2.0	2.1	1.9	3.3	6.0	6.0	5.6	5.0	3.9	4.6	2.3
	PrecMin	9.3	0.0	0.0	0.0	0.0	0.0	0.3	1.1	0.8	0.1	0.0	0.0	0.0
	PrecStd	4.7	0.6	0.6	0.4	0.6	0.8	1.3	1.3	1.1	1.5	1.2	1.0	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	78.3	82.3	88.1	95.0	99.6	95.1	93.1	90.9	86.2	79.1	72.4
		MCWB	48.6	50.7	52.8	55.7	58.7	61.4	61.8	63.5	62.0	59.8	54.1	49.2
	2%	DB	68.4	73.3	79.1	84.1	91.3	96.8	91.4	90.5	88.2	82.4	75.0	68.5
		MCWB	46.7	48.2	50.8	54.0	57.8	60.7	62.8	63.4	61.8	57.9	52.2	48.1
	5%	DB	64.3	70.4	75.3	81.6	89.4	93.0	90.2	88.4	84.4	80.9	72.4	64.4
		MCWB	45.2	47.4	49.6	52.9	57.1	60.7	63.0	63.2	61.2	57.1	51.3	45.7
	10%	DB	61.1	66.1	72.4	79.1	85.9	90.5	87.9	86.2	82.0	77.2	68.4	61.1
		MCWB	43.4	45.9	48.6	51.7	56.1	60.5	63.0	63.3	61.2	56.1	50.0	44.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	53.6	57.3	60.3	64.8	68.1	68.9	68.5	67.9	66.0	58.4	52.8
		MCDB	64.7	70.4	69.8	75.4	79.5	82.4	81.7	81.5	78.8	78.7	69.1	66.0
	2%	WB	49.0	50.6	54.1	57.8	62.7	66.4	67.4	67.3	66.4	63.4	55.5	49.8
		MCDB	63.9	67.9	70.8	72.8	78.2	81.2	80.3	80.7	77.5	75.3	69.1	64.1
	5%	WB	46.4	48.7	51.8	55.6	61.0	65.4	66.4	66.3	65.2	61.3	53.4	47.4
		MCDB	61.8	66.8	71.1	74.2	77.4	80.5	79.7	79.9	76.5	71.8	67.1	62.1
	10%	WB	44.1	46.7	50.0	53.5	59.5	64.3	65.5	65.5	64.1	59.3	51.1	44.8
		MCDB	60.0	64.7	69.4	73.6	76.5	80.0	79.2	79.1	75.6	69.6	66.0	59.4
(35) Mean Daily Temperature Range	5% DB	MDBR	32.1	32.7	33.9	34.3	33.1	30.8	26.4	25.5	26.2	29.8	31.4	31.0
		MCDBR	38.9	39.5	40.0	38.4	38.1	36.1	31.1	30.0	31.4	34.5	37.8	37.5
	5% WB	MCWBR	22.2	21.0	19.9	18.7	16.4	13.1	9.5	9.6	11.5	15.6	19.7	21.3
		MCWBR	35.1	35.0	35.7	34.0	32.9	28.9	24.8	24.4	23.7	26.6	31.7	33.2
(40) Clear-Sky Solar Irradiance	taub		0.245	0.251	0.264	0.295	0.311	0.339	0.364	0.344	0.325	0.282	0.261	0.247
		taud	2.694	2.664	2.614	2.501	2.493	2.472	2.438	2.515	2.526	2.658	2.650	2.670
	Ebn at Noon		322	327	327	316	309	298	291	297	301	311	312	316
		Edh at Noon	23	26	29	34	35	35	36	33	32	26	24	23
(44) All-Sky Solar Radiation	RadAvg		1192	1506	1898	2243	2385	2349	2123	2003	1786	1558	1268	1104
	RadStd		109	124	105	127	119	109	128	121	110	134	107	80

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185

Nomenclature: See separate page